

Math 241, F12
Quiz 12

Name: Answer
SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer. No calculator is allowed in the quiz.

Problem 1. (6 points)

(a) Change $(2, \pi/3, \pi/4)$ from spherical coordinates to rectangular coordinates.

$$\begin{aligned}x &= \rho \sin \phi \cos \theta = 2 \cdot \sin \frac{\pi}{4} \cdot \cos \frac{\pi}{3} = 2 \cdot \frac{\sqrt{2}}{2} \cdot \frac{1}{2} = \frac{\sqrt{2}}{2} \\y &= \rho \sin \phi \sin \theta = 2 \cdot \sin \frac{\pi}{4} \cdot \sin \frac{\pi}{3} = 2 \cdot \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{6}}{2} \\z &= \rho \cos \phi = 2 \cdot \cos \left(\frac{\pi}{4} \right) = 2 \cdot \frac{\sqrt{2}}{2} = \sqrt{2} \\&\Rightarrow \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{6}}{2}, \sqrt{2} \right)\end{aligned}$$

(b) Change $(1, \sqrt{3}, 2\sqrt{3})$ from rectangular coordinates to spherical coordinates.

$$\begin{aligned}\rho &= \sqrt{x^2 + y^2 + z^2} = \sqrt{1^2 + (\sqrt{3})^2 + (2\sqrt{3})^2} = \sqrt{16} = 4 \\ \cos \phi &= \frac{z}{\rho} = \frac{2\sqrt{3}}{4} = \frac{\sqrt{3}}{2} \Rightarrow \phi = \frac{\pi}{6} \\ \tan \theta &= \frac{y}{x} = \frac{\sqrt{3}}{1} = \sqrt{3} \\ &\quad (1, \sqrt{3}) \text{ is in the first quadrant} \Rightarrow \theta = \frac{\pi}{3} \\ &\Rightarrow \left(4, \frac{\pi}{3}, \frac{\pi}{6} \right)\end{aligned}$$

Problem 2. (4 points) Find the volume of the solid $E = \{(\rho, \theta, \phi) \mid 1 \leq \rho \leq 2, 0 \leq \theta \leq \pi/2, 0 \leq \phi \leq \pi/2\}$ using the triple integral in spherical coordinates.

$$\begin{aligned}V(E) &= \int_0^{\pi/2} \int_0^{\pi/2} \int_1^2 \rho^2 \sin \phi \, d\rho \, d\theta \, d\phi \\ &= \int_0^{\pi/2} \int_0^{\pi/2} \left[\frac{\rho^3}{3} \sin \phi \right]_{\rho=1}^{\rho=2} d\theta \, d\phi \\ &= \int_0^{\pi/2} \int_0^{\pi/2} \frac{7}{3} \sin \phi \, d\theta \, d\phi \\ &= \int_0^{\pi/2} \frac{7\pi}{6} \sin \phi \, d\phi \\ &= \frac{7\pi}{6} [-\cos \phi]_{\phi=0}^{\phi=\pi/2} \\ &= \frac{7\pi}{6}\end{aligned}$$